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SQL Cheat Sheet

Queries, joins, aggregations, window functions & schema design.

SELECT & FILTERING

```
SELECT col1, col2 FROM ta
SELECT * FROM table
SELECT DISTINCT col FROM
WHERE col > 10
WHERE col BETWEEN 1 AND 5
WHERE col IN ('a', 'b', '
WHERE col LIKE '%pattern%
```

SORTING & LIMITING

```
ORDER BY col ASC
ORDER BY col DESC
ORDER BY col1, col2 DESC
LIMIT 10
LIMIT 10 OFFSET 20

WHERE RANK() OVER(...) <=
```

JOINS

```
INNER JOIN t2 ON t1.id=t2
LEFT JOIN t2 ON t1.id=t2.
RIGHT JOIN t2 ON t1.id=t2
FULL OUTER JOIN t2 ON ...
CROSS JOIN t2-- cartesian
SELF JOIN: FROM t a JOIN
ON a.mgr_id = b.id
```

AGGREGATIONS

```
SELECT COUNT(*) FROM tabl
SELECT COUNT(DISTINCT col
SELECT SUM(col) FROM tabl
SELECT AVG(col) FROM tabl
SELECT MIN(col), MAX(col)
GROUP BY col1, col2
HAVING COUNT(*) > 5
```

SUBQUERIES & CTEs

```
WHERE id IN (SELECT id FR

WITH cte AS (
SELECT id, SUM(v) AS tota
FROM orders GROUP BY id)
SELECT * FROM cte WHERE t
```

WINDOW FUNCTIONS

```
ROW_NUMBER() OVER (ORDER
RANK() OVER (ORDER BY col
DENSE_RANK() OVER (ORDER
LAG(col, 1) OVER (ORDER B
LEAD(col, 1) OVER (ORDER
SUM(col) OVER (PARTITION
AVG(col) OVER (ORDER BY d
```

NULL HANDLING

```
WHERE col IS NULL
WHERE col IS NOT NULL
COALESCE(col, 0)-- first
NULLIF(col, 0)-- null if
ISNULL(col, 'default')--
IFNULL(col, 0)-- MySQL
col IS DISTINCT FROM othe
```

STRING FUNCTIONS

```
UPPER(col) / LOWER(col)
LENGTH(col) / LEN(col)
TRIM(col) / LTRIM / RTRIM
SUBSTRING(col, 1, 5)
CONCAT(col1, ' ', col2)
REPLACE(col, 'old', 'new'
col || ' suffix'-- concat
```

DATE FUNCTIONS

```
NOW() / CURRENT_TIMESTAMP
CURRENT_DATE / CURRENT_TI
DATE_TRUNC('month', col)
EXTRACT(YEAR FROM col)
DATEDIFF(day, start, end)
DATE_ADD(col, INTERVAL 7
STRFTIME('%Y-%m', col)--
```

INSERT, UPDATE, DELETE

```
INSERT INTO t (col1,col2)
VALUES ('a', 1)
INSERT INTO t SELECT * FR
UPDATE t SET col=val
WHERE condition
DELETE FROM t WHERE condi
TRUNCATE TABLE t-- fast d
```

SCHEMA & TABLES

```
CREATE TABLE t (
id INT PRIMARY KEY,
name VARCHAR(100) NOT NUL
val DECIMAL(10,2) DEFAULT
ALTER TABLE t ADD col INT
DROP TABLE t
CREATE INDEX idx ON t(col
```

EXPLAIN & OPTIMISE

```
EXPLAIN SELECT * FROM t
EXPLAIN ANALYZE SELECT ..
```